

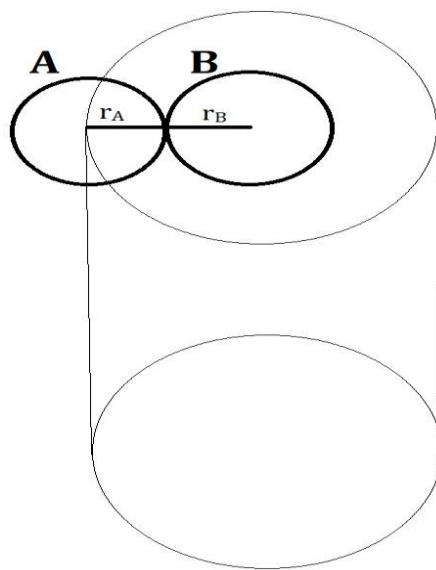
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Home Work – 1 Solutions



$$-\frac{dn_A}{dt} = Z_{AB} P \quad 1$$

Part 1



Swept area where collision occurs = $\pi (r_A + r_B)^2$

Moving volume where the collision occurs = $v_{AB} \pi (r_A + r_B)^2$

Only if at least 50 % of the individual molecule (A as well as B) comes in contact within this volume, the collision will occur else there will not be any collision.

The collision frequency is given as,

$$Z_{AB} = n_A n_B \pi (r_A + r_B)^2 \left(\frac{8kT}{\pi \mu} \right)^{\frac{1}{2}} \quad 2$$

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Part 2

$$-\frac{dn_A}{dt} = n_A n_B \pi(r_A + r_B)^2 \left(\frac{8kT}{\pi} \mu\right)^{\frac{1}{2}} \exp\left(-\frac{E^*}{k_B T}\right)$$

$C_A = n_A/N^A$ and $C_B = n_B/N^A$ where N^A is the Avogadro's number. Substituting in the above equation, we get rate equation as,

$$r = -\frac{dC_A}{dt} = \pi(r_A + r_B)^2 \left(\frac{8kT}{\pi} \mu\right)^{\frac{1}{2}} \exp\left(-\frac{E^*}{k_B T}\right) C_A C_B \quad 3$$

$$k = \pi(r_A + r_B)^2 \left(\frac{8kT}{\pi} \mu\right)^{\frac{1}{2}} \exp\left(-\frac{E^*}{k_B T}\right) \quad 4$$

Part 3

For complicated system, the rate constant can be extended by employing a **correction factor**, p which lies between 0 and 1.

$$k = p \pi(r_A + r_B)^2 \left(\frac{8kT}{\pi} \mu\right)^{\frac{1}{2}} \exp\left(-\frac{E^*}{k_B T}\right) \quad 5$$

Part 4

$$r_A = 165 \times 10^{-12} \text{ m}$$

$$r_B = 188 \times 10^{-12} \text{ m}$$

$$k_B = 1.3806 \times 10^{-23} \text{ m}^2 \text{ kg} / \text{s}^2 \text{ K}$$

$$\mu = m_A m_B / (m_A + m_B) = M_A M_B / N^A (M_A + M_B)$$

$$M_A = 46 \times 10^{-3} \text{ kg/mol}$$

$$M_B = 28.01 \times 10^{-3} \text{ kg/mol}$$

$$N^A = 6.022 \times 10^{23} / \text{mol}$$

$$E^* = 3 \times 10^{-20} \text{ J}$$

$$T = 300, 350, 400, 450, 500 \text{ K}$$

Substitute the above values in equation 5 to the value of the rate constant.

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T, K	k, (mol/m ³) ⁻¹ s ⁻¹
300	101800
350	309500
400	718900
450	1394400
500	2382300

MATLAB script for part 4

```
r_A = 165e-12; % Radius of NO2
r_B = 188e-12; % Radius of CO
k_B = 1.3806e-23; % Boltzmann Constant
M_A = 46e-3; % Molecular mass of NO2, kg/mol
M_B = 28.01e-3; % Molecular mass of CO, kg/mol
Na = 6.022e23; % Avogadro's number
E = 3e-20; % Energy to overcome electron cloud
repulsion, J
T = [300,350,400,450,500]; % Temperature, K
mu = (M_A*M_B)/((M_A + M_B)*Na) % Reduced mass

% Rate Constant
k = Na*pi*((r_A + r_B)^2)*(exp(-
E./(k_B.*T))).*(8.*k_B.*T./(pi*mu)).^(1/2)
```

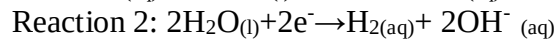
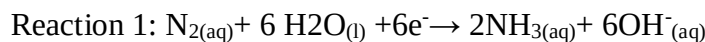
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Homework 2 - Solutions

A) Electrochemical Production of Ammonia (NH₃) and Hydrogen (H₂) in Alkaline Solutions [8 points]

In an electrochemical batch reactor with a total volume of 100 L, the following two irreversible reactions takes place at 20°C in alkaline solution.



The pH is kept constant throughout the reaction and H₂O can be considered in excess, such that reaction invariant is not dependent on the concentration of H₂O and pH. The species of interest are N₂, NH₃, and H₂ dissolved in water. Initially, the reaction mixture contains only N₂ at the solubility limit of 0.0019 gm/(100gm water). Assume concentration of H₂O to be 55 moles/liter. Solubility limit of hydrogen and ammonia is 0.00016 gr gas/(100 gr water) and 52.9 gm NH₃/100 mg water, respectively).

1) Determine the physical range of molar extents for reactions 1 and 2. What can be said about the rate of reaction 2 when the maximum molar extent is achieved? [Hint: sparingly soluble compounds can phase separate when their concentration increases beyond the saturation (or solubility) limit]

Solution:

Reaction 1:

$$N_{\text{N}_2} = 0$$

$$N_{\text{N}_{2,0}} = 0.0019 \frac{\text{gm gas}}{100 \text{ gm water}} \times \frac{1000 \text{ gm water}}{1 \text{ L water}} \times 100 \text{ L} \times \frac{1 \text{ mol}}{28 \text{ gm N}_2} = 0.06786 \text{ mol N}_2$$

$$X_1: \frac{N_{\text{N}_2} - N_{\text{N}_{2,0}}}{-1} = \frac{0 - 0.06786}{-1} = 0.06786 \text{ mol}$$

Reaction 2:

$$N_{\text{H}_{2,0}} = 0$$

$$N_{\text{H}_2} = 0.00016 \frac{\text{gm gas}}{100 \text{ gm water}} \times \frac{1000 \text{ gm water}}{1 \text{ L water}} \times 100 \text{ L} \times \frac{1 \text{ mol}}{2 \text{ gm H}_2} = 0.08 \text{ mol H}_2$$

$$X_1: \frac{N_{\text{H}_2} - N_{\text{H}_{2,0}}}{+1} = \frac{0.08 - 0}{+1} = 0.08 \text{ mol}$$

The reaction continues at the same rate but instead of producing aqueous hydrogen it produces gaseous hydrogen.

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2) Find the reaction invariant for the species involved in reactions 1 and 2, and plot it using Matlab for the physical range of molar extents determined above.

Solution:

$$N_{NH_3,0} = 0$$

$$\text{from reaction 1: } \frac{N_{N_2} - N_{N_2,0}}{-1} = \frac{N_{NH_3} - N_{NH_3,0}}{2} = 0.06786 \text{ mol}$$

$$N_{NH_3} = 0.13572 \text{ mol}$$

From solubility limit:

$$N_{NH_3, \text{limit}} = 52.9 \frac{\text{gm gas}}{100 \text{ mg water}} \times \frac{1000 \text{ g water}}{1 \text{ L water}} \times 100 \text{ L} \times \frac{1 \text{ mol}}{17 \text{ gm } NH_3} = 3111 \text{ mol}$$

The resulted value is lower than solubility limit therefore is acceptable

$$A_1 = N_2 \quad A_2 = NH_3 \quad A_3 = H_2$$

$$\alpha = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$NA - NA,0 = \begin{bmatrix} N_{N_2} - N_{N_2,0} \\ N_{NH_3} - N_{NH_3,0} \\ N_{H_2} - N_{H_2,0} \end{bmatrix}$$

$$N - N_0 = \alpha^T * X$$

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N 1	0.067 86	0.067 86	0.067 86	0.067 86	0.067 86	0.050 895	0.050 895	0.050 895	0.050 895	0.050 895	0.033 93	0.033 93	0.033 93	0.033 93	0.033 93	0.016 965	0.016 965	0.016 965	0.016 965	0.016 965	0	0	0	0	0
N 2	0	0	0	0	0	0.033 93	0.033 93	0.033 93	0.033 93	0.033 93	0.067 86	0.067 86	0.067 86	0.067 86	0.067 86	0.101 79	0.101 79	0.101 79	0.101 79	0.101 79	0.135 72	0.135 72	0.135 72	0.135 72	0.135 72
N 3	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08

```

% CHE 321 Chemical Reaction engineering
% Home work 2
%-----
%-----Part A-----
% Calculating X
clear
clc
alpha=[-1 2 0
        0 0 1];
%X= (alpha*alpha')*alpha*(N)
%N= N-N0

%-----
% plotting X
% N=N-N0

%X=A*(N)
x1=linspace(0,0.06786,5);
x2=linspace(0,0.08,5);
X= [x1
    x2];

count=0;
x=zeros(2,1);

for i=1:length(X)
    n1=X(1,i);
    for m=1:length(X)
        n2=X(2,m);
        x=[n1;n2];
        count = count+1;
        N(:,count)= (alpha')*x;
    end
end
for i=1:length(N)
    N(1,i)=N(1,i)+ 0.06786
end

```

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3) *How does the reaction invariant change when H_2 is also present initially at the solubility limit of 0.00016 gm gas/(100 gm water)?*

Solution:

The reaction invariant for the first reaction doesn't change

But for the second reaction this value would be equal to zero because after reaching the solubility limit the reaction produce hydrogen in gas form instead of aqueous form.

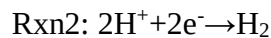
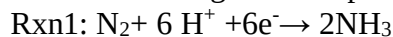
B) Electrochemical Production of Ammonia (NH_3) and Hydrogen (H_2) in Acidic Solutions [8 points]

Now consider the previous reactions 1 and 2 in a semi batch reactor with a constant supply of Nitrogen gas to an acidic solution, such that the solution is saturated with N_2 throughout the course of reactions 1 & 2. Here the species of interest are H^+ , NH_3 , and H_2 dissolved in water. Initially, there are no products and the pH is 1.

1) *Write the reactions 1 & 2 of acidic solution involving H^+ ions.*

Solution

since the pH is changing the reaction should change to incorporate H^+ .



2) *Find the reaction invariant for the species in the new reactions and plot it in Matlab for a physical range of molar extents.*

solution

$$A_1 = H^+, A_2 = NH_3, A_3 = H_2$$

$$pH = -\log[H^+] = 1$$

$$[H^+] = 10^{-1} = 0.1$$

$$N_{H^+} = 0.1 * 100 = 10 \text{ mol}$$

$$X_2: \frac{N_{H_2} - N_{H_2,0}}{1} = \frac{N_{H^+} - N_{H^+,0}}{-2} = \frac{0 - 10}{-2} = 5$$

But since this value is greater than the solubility limit then

$$N_{H_2, \text{limit}} = 0.00016 \frac{\text{gm gas}}{100 \text{ gm water}} \times \frac{1000 \text{ gm water}}{1 \text{ L water}} \times 100 \text{ L} \times \frac{1 \text{ mol}}{2 \text{ gm } H_2} = 0.08 \text{ mol } H_2$$

$$X_2: \frac{N_{H_2} - N_{H_2,0}}{1} = \frac{0.08 - 0}{1} = 0.08$$

$$X_1: \frac{N_{H^+} - N_{H^+,0}}{-6} = \frac{N_{NH_3} - N_{NH_3,0}}{2} = \frac{0 - 10}{-6} = 1.667$$

$$N_{NH_3} = 3.333$$

$$\alpha = \begin{bmatrix} -6 & 2 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$NA-NA,0=\begin{bmatrix} N_{H^+}-N_{H^+_0} \\ N_{NH_3}-N_{NH_3} \\ N_{H_2}-N_{H_{2,0}} \end{bmatrix}$$

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N1	10	9.96	9.92	9.88	9.84	7.5	7.46	7.42	7.38	7.34	5	4.96	4.92	4.88	4.84	2.5	2.46	2.42	2.38	2.34	-2.0007
N2	0	0	0	0	0	0.833333	0.833333	0.833333	0.833333	0.833333	1.666667	1.666667	1.666667	1.666667	1.666667	2.5	2.5	2.5	2.5	2.5	3.333333
N3	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08	0	0.02	0.04	0.06	0.08	0

```
%-----
%-----Part B-----
alpha2=[-6 2 0
        -2 0 1];
NHplus0=10; %NNH30=0 , NH20=0;
% plotting X
% N=N-N0
m1=linspace(0.0,1.6666667,5);
m2=linspace(0,0.08,5);
X2= [m1;
     m2];

count=0;
x=zeros(2,1);
%
for i=1:length(X2)
    n1=X2(1,i);
    for m=1:length(X2)
        n2=X2(2,m);
        x=[n1;n2];
        count = count+1;
        N2(:,count)= (alpha2')*x;
        if count> 100
            break;
        end
    end
end
for i=1:length(N2)
    N2(1,i)=N2(1,i)+ NHplus0;
end
```

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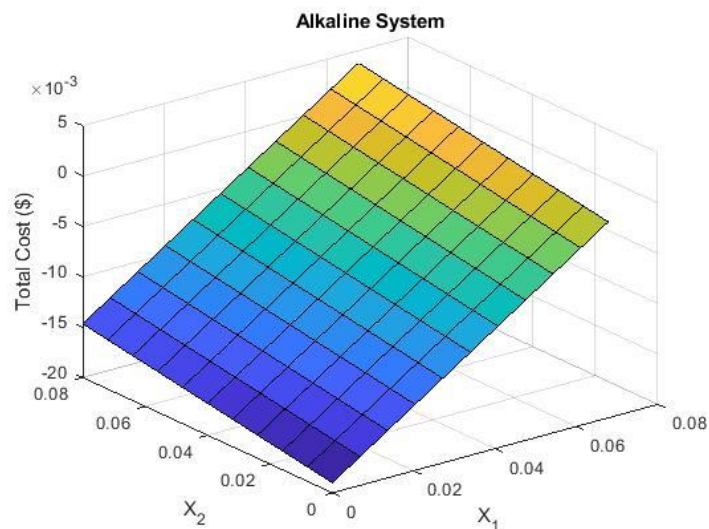
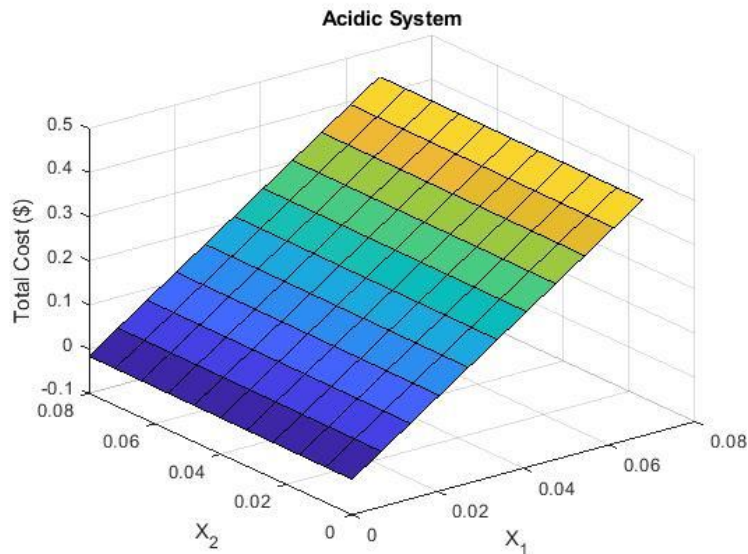
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A) Economic Analysis of Ammonia and Hydrogen Production [4 points]

Consider the cost of N_2 to be 10 \$/kg (70 \$/40L in 135 bar cylinder), NH_3 to be 8 \$/kg, H_2 to be 27 \$/kg (500\$ in 300 bar 50L cylinder) and HCl to be 0.09 \$/kg (25 \$ per 270 kg).

1) Obtain the total material cost for the range of molar extents for reactions in alkaline and acidic solution and discuss the economic feasibility of the NH_3 production in alkaline vs acidic solution.

Solution:



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Total profit for the acidic system is higher as NH₃ production and N₂ utilization is much greater in the acidic system than it is in alkaline. The acidic mixture is better in terms of NH₃ production and for some cases in the alkaline system the total cost value is negative. That means the cost of providing nitrogen is much higher than the profit made from the NH₃ and H₂.

Note that we are keeping the mixture saturated with nitrogen in the acidic system. So the nitrogen continuously should be added to the mixture. So the amount of needed N₂ is much higher than the nitrogen saturation value. This amount depends on the reaction rate and the operation time of the reactor. Also, it is reasonable to assume that extra amount of the produced hydrogen (the produced hydrogen in gas form after the solution reaches its solubility limit) is going to be captured and sold. Again, for calculating this value in part A, reaction rate and operation time are required.

Considering the facts mentioned above it cannot conclusively be reported that the second reaction is economically favorable although it seems promising. For a conclusive result, a deeper economic calculation should be conducted.

-----Part C-----

```
N2Price=10;
NH3Price=8
H2Price=27
HCLPrice=0.09
N0N2=ones(1,length(N))*0.06782;
N0Hplus=ones(1,length(N))*5;
TCost_1=-N0N2(1,:)*28*N2Price/1000+N(2,:)*17*NH3Price/1000+N(3,:)*2*H2Price/1000
TCost_2=-N0N2(1,:)*28*N2Price/1000-
N0Hplus(1,:)*1*HCLPrice/1000+N2(2,:)*17*NH3Price/1000+N2(3,:)*2*H2Price/1000

[X1,X2]=meshgrid(0:0.006782:0.06782,0:0.008:0.08)
TCost1=-
(alpha(1,1)*X1+alpha(1,2)*X2)*28*N2Price/1000+(alpha(1,2)*X1+alpha(2,2)*X2)*17*NH3Price/1000+(alpha(1,3)*X1+alpha(2,3)*X2)*2*H2Price/1000
figure(01)
surf(X1,X2,TCost1)
title('Alkaline System')
xlabel('X_1')
ylabel('X_2')
zlabel('Total Cost ($)')
[M1,M2]=meshgrid(0:0.333:3.33,0:0.008:0.08)
TCost2=-
(alpha2(1,1)*M1+alpha2(1,2)*M2)*HCLPrice/1000+(alpha2(1,2)*M1+alpha2(2,2)*M2)*17*NH3Price/1000+(alpha2(1,3)*M1+alpha2(2,3)*M2)*2*H2Price/1000
figure(02)
```

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```
surf(X1,X2,TCost2)
```

```
title('Acidic System')
```

```
xlabel('X_1')
```

```
ylabel('X_2')
```

```
zlabel('Total Cost ($)')
```

```
% x=1:1:length(TCost_1);
```

```
% x2=1:1:length(TCost_2);
```

```
% figure(03)
```

```
% plotyy(x,TCost_1,x2,TCost_2);
```

```
% hold on;
```

Problem 3:

1. (2 points)



2. (4 points)

$$n_{\text{Cu}} = \frac{n_{\text{H}_2\text{SO}_4}}{2} = n_{\text{SO}_2} = \left(\frac{1000g}{64 \frac{g}{\text{mol}}} \right) = 15.625 \text{ mol}$$

$$m_{\text{Cu}} = 15.625 \text{ mol} \times 64 \frac{g}{\text{mol}} = 1000 \text{ g} = 1 \text{ kg}$$

$$m_{\text{H}_2\text{SO}_4} = 2 \times 15.625 \text{ mol} \times 98 \frac{g}{\text{mol}} = 3062.5 \text{ g}$$

$$m_{\text{acid solution}} = \frac{3062.5 \times 100}{94} = 3258 \text{ g} = 3.258 \text{ kg}$$

3. (4 points)

$$n_{\text{H}_2\text{SO}_4} = \frac{15,000 \text{ g} \times 94\%}{98 \frac{g}{\text{mol}}} = 143.88 \text{ mol}$$

$$n_{\text{Cu}} = \frac{1000 \text{ g}}{64 \frac{g}{\text{mol}}} = 15.625 \text{ mol}$$

$$n_{\text{acid left}} = 143.88 - 15.625 \times 2 = 112.63 \text{ mol}$$

$$m_{\text{acid left}} = 112.63 \text{ mol} \times 98 \frac{g}{\text{mol}} = 11037.74 \text{ g} = 11.038 \text{ kg}$$

$$m_{\text{solution}} = 1 \text{ kg} + 15 \text{ kg} - 15.625 \text{ mol} \times 0.064 \frac{\text{kg}}{\text{mol}} = 15 \text{ kg}$$

$$\% \text{H}_2\text{SO}_4 = \frac{11.038 \text{ kg}}{15 \text{ kg}} \times 100 = 73.59\%$$